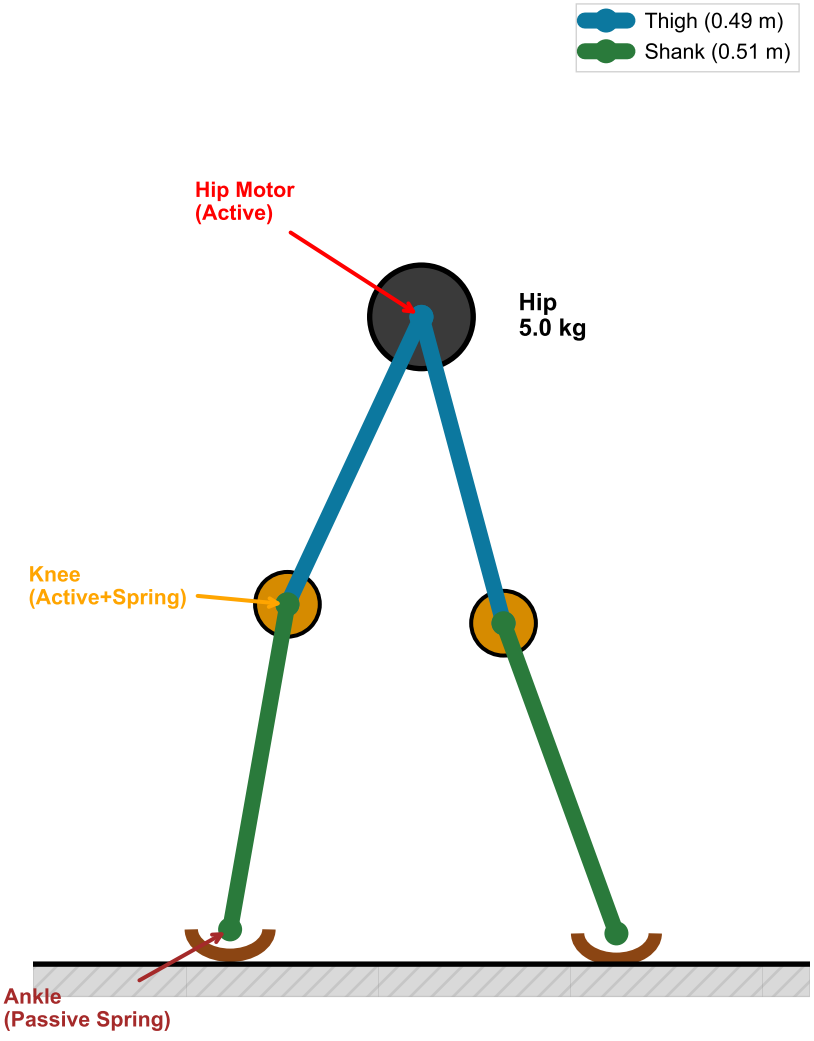
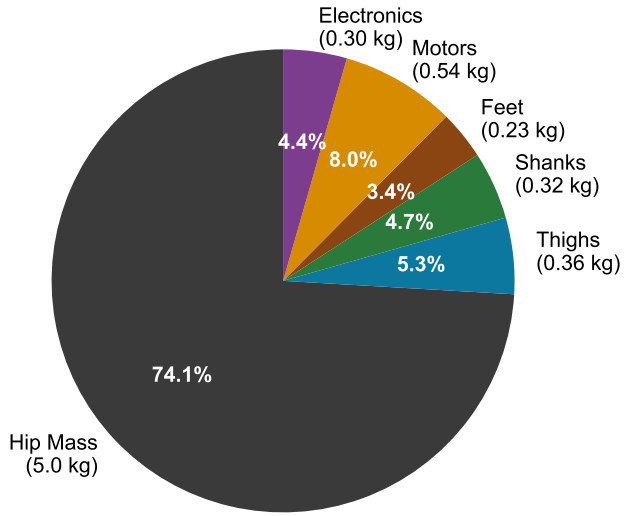
V22 Hardware Design: Mechanical Structure and Specifications

(A) Robot Mechanical Design



(B) Mass Distribution
(Total: 6.75 kg)



(C) Hardware Specifications (V22 Design)

Component	Specification	Value
Motor Model	Dynamixel MX-64AR	4 units
Peak Torque	Rated / Stall	6.0 / 7.3 N·m
Software Limit	Safe Operating	4.0 N·m
Encoder Res.	Position Feedback	4096 CPR
Hip Stiffness	Active Joint	Variable
Knee Stiffness	Active + Spring	0.8 N·m/rad
Ankle Stiffness	Passive Spring	10.0 N·m/rad
Control Freq.	Main Loop	1000 Hz
Battery	3S LiPo	11.1V 2.2Ah
Leg Length	Thigh + Shank	1.00 m